Appendix: Experimental Results and Data

Scalable Quantum Error Decoding with Sparse Graph Neural Networks

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1 Phase 1 — Architecture Selection

Table A1. Final validation accuracy and logical error rate for each GNN architecture at d = 7 (10 epochs). MWPM included as reference.

Architecture	Val. Accuracy	Acc. Std	Logical Error Rate	LER Std	Final Loss
GCN	0.8148	0.0043	0.1852	0.0043	—
GAT	0.8157	0.0059	0.1843	0.0059	—
GIN	0.8817	0.0193	0.1183	0.0193	—
GraphSAGE	0.8923	0.0005	0.1077	0.0005	—
MWPM	0.9849	0.0000	0.0151	N/A	N/A

Architecture ranking by accuracy: (1) GraphSAGE (0.8923), (2) GIN (0.8817), (3) GAT (0.8157), (4) GCN (0.8148).

Selected architecture: GraphSAGE — best accuracy, lowest logical error rate, and lowest variance (std = 0.0005) among GNN models.

2 Phase 2 — Hyperparameter Tuning

Table A2. All 25 random search configurations for GraphSAGE with resulting validation and test accuracy. Best configuration (ID 13) highlighted.

ID	Layers	Hidden	LR	Dropout	Aggr	Val Acc	Test Acc	Loss	Time (s)
0	2	64	0.0005	0.1	mean	0.8626	0.8608	0.2818	3132
1	3	64	0.0030	0	max	0.9343	0.9343	0.1327	3302
2	2	128	0.0030	0.3	mean	0.8746	0.8744	0.2727	5356
3	4	256	0.0003	0.1	max	0.9285	0.9282	0.1488	16863
4	2	64	0.0010	0	max	0.9035	0.9035	0.1966	2389
5	4	256	0.0001	0.3	mean	0.8903	0.8896	0.2431	15453
6	5	64	0.0030	0.2	max	0.8919	0.8907	0.1999	8891
7	5	256	0.0003	0.2	max	0.9057	0.9048	0.1537	21998
8	3	256	0.0001	0.1	mean	0.8837	0.8829	0.2432	9725
9	3	512	0.0010	0.2	mean	0.9081	0.9085	0.1912	29680
10	5	256	0.0001	0.1	max	0.9266	0.9258	0.1444	20649
11	3	512	0.0010	0.3	mean	0.9050	0.9047	0.2115	23621
12	4	128	0.0003	0.2	max	0.8935	0.8934	0.1971	10067
13	**5**	**256**	**0.0003**	**0.1**	**max**	**0.9390**	**0.9388**	**0.1268**	**29305**
14	4	64	0.0001	0.2	max	0.8691	0.8681	0.2427	6178
15	2	256	0.0010	0.1	max	0.8971	0.8961	0.2109	9068
16	2	256	0.0010	0	mean	0.8876	0.8881	0.2095	3520
17	4	256	0.0003	0	mean	0.9083	0.9101	0.1092	5788
18	2	64	0.0010	0	mean	0.8754	0.8765	0.2434	1986
19	5	256	0.0010	0.3	mean	0.9259	0.9253	0.1856	19076
20	3	128	0.0001	0.2	mean	0.8703	0.8694	0.2763	6251
21	3	128	0.0001	0	mean	0.8788	0.8792	0.2359	3280
22	2	64	0.0010	0.2	max	0.8745	0.8730	0.2668	3513
23	5	512	0.0001	0	mean	0.9179	0.9186	0.0660	23490
24	2	64	0.0030	0	mean	0.8878	0.8881	0.2202	1975

Best Configuration (ID 13)

Layers: 5, Hidden dim: 256, Learning rate: 0.0003, Dropout: 0.1, Aggregation: max

Validation accuracy: 0.9390, Test accuracy: 0.9388, Final loss: 0.1268

Batch size: 256, Epochs: 50, Training time: 29,305 s (~8.1 h)

3 Phase 3 — Benchmarking

3.1 Accuracy and Logical Error Rate

Table A3. Per-distance final accuracy, logical error rate, and loss for Specialist GSAGE vs Specialist MLP (50 epochs).

d	Model	Accuracy	Logical Error Rate	Final Loss	Train / Val Acc
3	GSAGE	0.9869	0.0131	0.0384	0.9874 / 0.9898
3	MLP	0.9872	0.0128	0.0376	0.9883 / 0.9872
5	GSAGE	0.9799	0.0201	0.0505	0.9798 / 0.9806
5	MLP	0.9774	0.0226	0.0452	0.9862 / 0.9774
7	GSAGE	0.9484	0.0516	0.1225	0.9435 / 0.9491
7	MLP	0.9166	0.0835	0.1497	0.9375 / 0.9166
9	GSAGE	0.8886	0.1114	0.2201	0.8946 / 0.8886
9	MLP	0.7913	0.2087	0.4882	0.8048 / 0.7913
11	GSAGE	0.8345	0.1655	0.3065	0.8298 / 0.8345
11	MLP	0.6292	0.3708	0.5614	0.6279 /0.6292
13	GSAGE	0.7427	0.2573	0.3761	0.7775 / 0.7427
13	MLP	0.5759	0.4241	0.6288	0.5771 / 0.5759

3.2 Parameter Counts

Table A4. Model parameter counts and size (MB) for GNN vs MLP at each code distance. Param ratio = GNN / MLP.

d	MLP Params	GNN Params	MLP (MB)	GNN (MB)	Ratio
3	2,646,529	547,201	30.31	2.11	0.207
5	2,695,681	547,201	30.87	2.11	0.203
7	2,806,273	547,201	32.14	2.11	0.195
11	3,310,081	547,201	37.90	6.30	0.165
13	3,752,449	547,201	42.96	6.30	0.146

3.3 Inference Time

Table A5. Inference latency (μs per sample) and throughput (samples/s) for GNN, MLP (NN), and MWPM at each distance.

d	Model	Latency (μs)	Throughput (samples/s)	Batch
3	MLP	4.12	242,506	64
3	GNN	40.64	24,606	64
3	MWPM	0.11	9,080,546	—
5	MLP	4.03	247,855	64
5	GNN	41.14	24,310	64
5	MWPM	0.98	1,025,482	—

d	Model	Latency (µs)	Throughput (samples/s)	Batch
7	MLP	4.74	210,783	64
7	GNN	43.20	23,149	64
7	MWPM	3.42	292,284	—
9	MLP	4.76	210,148	64
9	GNN	44.35	22,550	64
9	MWPM	8.30	120,548	—
11	MLP	4.90	204,164	64
11	GNN	40.65	24,598	64
11	MWPM	16.07	62,235	—
13	MLP	5.05	197,995	64
13	GNN	50.42	19,834	64
13	MWPM	27.91	35,831	—

3.4 Training Curve (d = 7 Specialist GSAGE)

Table A6. Per-epoch training loss, training accuracy, and validation accuracy for Specialist GSAGE at d = 7.

Epoch	Train Loss	Train Accuracy	Val Accuracy
1	0.3114	0.8392	0.8542
2	0.2412	0.8772	0.8728
3	0.2197	0.8891	0.8835
4	0.2053	0.8975	0.8907
5	0.1946	0.9040	0.9014
6	0.1866	0.9086	0.8946
7	0.1798	0.9127	0.9131
8	0.1747	0.9156	0.9158
9	0.1704	0.9181	0.9204
10	0.1661	0.9202	0.9263
25	0.1392	0.9348	0.9414
50	0.1225	0.9435	0.9491

4 Phase 4 — Extrapolation

4.1 Extrapolation Accuracy by Split

Table A7. Mean accuracy (%) on unseen evaluation distances for all 15 training-split experiments. Trained on d = 3, 5, 7; evaluated on d = 9, 11, 13.

Split	d3 %	d5 %	d7 %	Acc. d = 9	Acc. d = 11	Acc. d = 13
equal_333333	33	33	34	78.68	66.35	58.18
a1_d3_00	0	50	50	78.88	66.56	57.73
a2_d3_10	10	45	45	79.31	66.26	57.45
a3_d3_20	20	40	40	79.48	66.82	57.82
a4_d3_40	40	30	30	79.34	67.75	59.21
a5_d3_50	50	25	25	79.45	68.54	60.74
b1_d5heavy	0	80	20	80.54	67.76	59.19
b2_d5more	0	65	35	80.13	66.66	58.03
b3_balanced	0	50	50	80.68	68.73	59.62
b4_d7more	0	35	65	79.49	67.13	58.21
b5_d7heavy	0	20	80	77.57	64.75	56.77
c1_only_d3	100	0	0	63.15	58.22	55.15
c2_only_d5	0	100	0	68.13	59.65	55.47
c3_only_d7	0	0	100	76.54	63.05	55.46
c4_no_d7	50	50	0	66.33	59.21	55.37

4.2 Training Summary by Split

Table A8. Training, validation, and test accuracy on the training distances, with final loss and wall-clock time for each split.

Split	Train Acc %	Val Acc %	Test Acc %	Final Loss	Time (min)
a1_d3_00	96.24	96.28	96.14	0.0907	4.3
a2_d3_10	96.24	96.48	96.35	0.0869	6.4
a3_d3_20	96.79	96.57	96.61	0.0833	208.1
a4_d3_40	97.05	96.92	96.92	0.0751	331.1
a5_d3_50	97.22	97.51	97.28	0.0704	9.4
b1_d5heavy	97.43	97.04	97.18	0.0678	5.6
b2_d5more	96.49	96.72	96.68	0.0794	5.6
b3_balanced	95.99	96.53	96.14	0.0920	35.0
b4_d7more	95.92	95.59	95.83	0.1013	731.1
b5_d7heavy	95.40	95.26	95.26	0.1108	801.7
c4_no_d7	98.47	98.16	98.20	0.0480	243.7
equal_333333	96.69	96.67	96.59	0.0783	7.6

Note: c1_only_d3, c2_only_d5, c3_only_d7 training logs were not saved.

4.3 Wilcoxon Signed-Rank Test Results

*Table A9. Paired comparisons of extrapolation accuracy across all (code distance, physical error rate) configurations. Significance: *** p < 0.001, ** p < 0.01, * p < 0.05, n.s. = not significant.*

Exp	Comparison	$\Delta\bar{x}$ (%)	p-value	Sig.	Interpretation
A	Control vs. No d = 3	**+0.01%**	0.977	n.s.	Dropping d = 3 does not hurt extrapolation
A	Control vs. 10% d = 3	**+0.06%**	0.741	n.s.	10% d = 3 effectively same as control
A	No d = 3 vs. 50% d = 3	**−1.85%**	0.0005	***	50% d = 3 is clearly better than 0% d = 3
B	d5-heavy vs. d7-heavy	**+2.80%**	0.0001	***	d5-heavy beats d7-heavy by ~3%
B	Balanced vs. d7-heavy	**+3.31%**	< 0.0001	***	Balanced (50/50) clearly better than d7-heavy
C	Only d = 7 vs. Only d = 3	**+6.18%**	0.0001	***	Only d = 7 extrapolates far better than only d = 3
C	Control vs. Only d = 7	**+2.72%**	< 0.0001	***	Balanced mix beats even the best single distance
C	Control vs. Only d = 3	**+8.90%**	< 0.0001	***	Control beats only-d = 3 by ~9%
A–B	No d = 3 (a1) vs. d7-heavy (b5)	**+1.36%**	0.0003	***	No-d = 3 training beats d7-heavy

Key Takeaways

1. Multi-distance training decisively outperforms single-distance. The balanced control beats only-d = 3 by +8.9% and only-d = 7 by +2.7% (both $p < 0.0001$).

2. Favour balanced or d = 5-heavy splits over d = 7-heavy. Balanced outperforms d7-heavy by +3.3% ($p < 0.0001$).

3. d = 3 helps as an additive, but not as a substitute. Including 50% d = 3 improves over 0% d = 3 by +1.9% ($p = 0.0005$), yet only-d = 3 is the worst configuration.

Method: Wilcoxon signed-rank test on paired extrapolation accuracies. Each pair compares two training compositions at the same (d, p) point. All p-values are two-sided. The test is non-parametric and does not assume normality.

4.4 DeepSet Baseline Comparison

Table A10. GNN (GraphSAGE) vs DeepSet accuracy and F1 score at each evaluation distance (equal_333333 model).

d	DeepSet Acc	GNN Acc	DeepSet F1	GNN F1	Δ Accuracy
3	0.9181	0.9837	0.5545	0.8968	+0.0656
5	0.9212	0.9734	0.8051	0.9245	+0.0522
7	0.8219	0.9117	0.6657	0.8284	+0.0898
9	0.5499	0.7951	0.3141	0.6981	+0.2452
11	0.5349	0.6777	0.3170	0.5931	+0.1428
13	0.5257	0.6078	0.3065	0.5338	+0.0821

Table A11. Wilcoxon signed-rank test: GNN vs DeepSet per distance. All comparisons significant ($p < 10^{-11}$).

d	W Statistic	p-value	Significant	Cohen's d	Δ Mean
3	0.0	1.59×10^{-11}	Yes	−1.364	+0.0656
5	0.0	1.58×10^{-11}	Yes	−0.698	+0.0522
7	0.0	1.59×10^{-11}	Yes	−1.035	+0.0898
9	0.0	1.59×10^{-11}	Yes	−5.459	+0.2452
11	0.0	1.59×10^{-11}	Yes	−4.598	+0.1428
13	0.0	1.59×10^{-11}	Yes	−3.476	+0.0821

GNN (GraphSAGE) significantly outperforms DeepSet at all distances, with the gap widening dramatically on unseen distances ($d = 9$: +24.5%, $d = 11$: +14.3%). This confirms that the graph-structured inductive bias — not merely permutation invariance — is responsible for the GNN's extrapolation ability.

4.5 Scaling Exponents

Table A12. Fitted error scaling exponents (α) vs theoretical values, with R^2 and log C, for selected splits.

Split	d	Fitted α	Theoretical α	Δα	R^2	log C
a1_d3_00	9	1.934	5.0	−3.066	0.975	8.745
a1_d3_00	11	1.172	6.0	−4.828	0.899	5.234
a1_d3_00	13	0.591	7.0	−6.409	0.770	2.366
b3_balanced	9	2.139	5.0	−2.861	0.976	9.714
b3_balanced	11	1.341	6.0	−4.659	0.907	6.056
b3_balanced	13	0.720	7.0	−6.280	0.821	3.014
equal_333333	9	2.018	5.0	−2.982	0.969	9.194
equal_333333	11	1.251	6.0	−4.749	0.880	5.654
equal_333333	13	0.611	7.0	−6.389	0.781	2.463

5 Dataset and Training Details

5.1 Sample Counts

Table A14. Total sample counts and train/validation/test split ratios by experimental phase.

Phase	Total Samples	Train Ratio	Val Samples	Test Ratio	Test Total	Notes
Phase 3	1,000,000	80%	10,000	20%	200,000	Per distance
Phase 4	1,000,000	80%	10,000	20%	200,000	Mixed d = 3, 5, 7

5.2 Training Hyperparameters

Table A15. Training hyperparameters used in each experimental phase.

Phase	Epochs	Batch	LR	Hidden	Layers	Dropout	Aggr	p values
Phase 1	10	—	—	128	4	—	—	0.001–0.007
Phase 2	50	256	varied	varied	varied	varied	varied	0.001–0.007
Phase 3	50	256	0.0003	256	5	0.1	max	0.001–0.007
Phase 4	50	256	0.0003	256	5	0.1	max	0.001–0.007

Physical error rates: $p \in \{0.001, 0.003, 0.005, 0.007\}$ with equal weights (0.25 each) for training.

Extrapolation evaluation: $p \in \{0.001, 0.002, 0.003, 0.004, 0.005, 0.006, 0.007, 0.008\}$, with 10,000 samples per (p, d) pair.

5.3 Graph Statistics

Table A16. Mean sparse graph statistics per code distance at p = 0.005. Sparsity = (mean nodes / detectors) × 100%.

d	Detectors	Avg Nodes	Std Nodes	Avg Edges	Std Edges	Sparsity %
3	24	1.4	1.6	3.0	6.2	5.8
5	120	8.4	4.3	47.1	29.1	7.0
7	336	24.7	7.4	148.4	44.3	7.4
9	720	55.4	11.1	332.3	66.5	7.7
11	1320	104.6	15.0	627.3	90.0	7.9
13	2184	176.8	20.2	1060.8	121.0	8.1

6 Surface Code Circuit